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MULTI-USE APPARATUS FOR AUDIO EQUIPMENT

Abstract

A multi-use apparatus includes a rectangular block fabricated from a foam having a closed-cell structure. The rectangular block includes a first face, a second face opposing the first face, and four side faces extending between the first face and the second face. At least two of the side faces have dimensions varying in height, width, or length. The rectangular block is configured to support a lower end of a sound projection device at an angle relative to a surface, the rectangular block providing acoustic isolation and/or vibration dampening between the sound projection device and the surface. Positioning the rectangular block between a proximal and a distal edge of the lower end of the sound projection device selectively adjusts the angle of the sound projection device relative to the surface, thereby modifying a direction of sound projection.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application claims the benefit of and priority to U.S. Provisional Patent Application No. 63/558,018, filed on Feb. 26, 2024, the entire contents of which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to a multi-use apparatus, and, more specifically, to a multi-use apparatus configured for use with audio equipment.

BACKGROUND

[0003] In the music industry, there is a frequent need for audio equipment that can provide support and/or acoustic isolation for sound projection devices while also accommodating the storage, transportation, and/or maintenance of equipment such as musical instruments. Conventional solutions often require separate stands or wedges for tilting sound projection devices and separate neck supports or padding for instruments. These disparate products can be cumbersome, unable to adequately dampen vibrations, and fail to provide sufficient protection during transport or maintenance. For example, wedge-type devices can tilt a speaker cabinet but offer minimal or no vibration dampening. Moreover, such devices are limited to a single tilt angle, and are otherwise too bulky for convenient transport. Metal-framed stands may provide limited adjustability, but can transfer low-frequency vibrations directly through the frame to the floor or stage. Other accessories, such as foam neck supports, are designed strictly to cradle and protect a musical instrument's headstock and neck during string changes or repairs and thus fail to address speaker tilt or acoustic isolation at all. As a result, musicians frequently purchase and carry multiple items, leading to added expense, clutter, and logistical challenges.

[0004] Accordingly, there is a need for a multi-use apparatus that addresses these challenges by offering an easy-to-use, durable, and/or adaptable apparatus for use with audio equipment.

SUMMARY

[0005] In accordance with an aspect of the present disclosure, a multi-use apparatus includes a rectangular block fabricated from a foam having a closed-cell structure. The rectangular block includes a first face, a second face opposing the first face, and four side faces extending between the first face and the second face. At least two of the side faces have dimensions varying in height, width, or length. The rectangular block is configured to support a lower end of a sound projection device at an angle relative to a surface, the rectangular block providing acoustic isolation and/or vibration dampening between the sound projection device and the surface. Positioning the rectangular block between a proximal and a distal edge of the lower end of the sound projection device selectively adjusts the angle of the sound projection device relative to the surface, thereby modifying a direction of sound projection.

[0006] In an aspect of the present disclosure, the rectangular block may be further configured to support a neck of a musical instrument above the surface during maintenance of the musical instrument.

[0007] In another aspect of the present disclosure, the rectangular block may be further configured to secure a position of a musical instrument within a case during storage and/or transportation of the musical instrument.

[0008] In yet another aspect of the present disclosure, the rectangular block may be further configured to provide a space between a plurality of musical instruments stored on a stand or a rack.

[0009] In a further aspect of the present disclosure, the rectangular block may be fabricated from cross-linked polyethylene (XLPE), chemically cross-linked polyethylene (XPE), and/or ethylene vinyl acetate (EVA).

[0010] In yet a further aspect of the present disclosure, the rectangular block may have a length ranging from approximately 2 inches to 6 inches, a width ranging from approximately 1.5 inches to 4 inches, and a height ranging from approximately 1 inch to 3 inches.

[0011] In another aspect of the present disclosure, the rectangular block may have a first orientation configured to tilt the sound projection device at a first angle relative to the surface, and a second orientation configured to tilt the sound projection device at a second angle relative to the surface. The second orientation may be obtained by rotating the rectangular block about an axis relative to the surface.

[0012] In yet another aspect of the present disclosure, the first angle may range from approximately 15 degrees to 31 degrees, and the second angle may range from approximately 25 degrees to 33 degrees.

[0013] In a further aspect of the present disclosure, the rectangular block may further include at least one arcuate recess defined in a first side face thereof. The arcuate recess may be configured to accommodate a neck of a musical instrument.

[0014] In yet a further aspect of the present disclosure, the rectangular block may further include an external layer configured to enhance a grip between the rectangular block and the surface.

[0015] In an aspect of the present disclosure, a method for using a multi-use apparatus includes: providing a rectangular block fabricated from a foam having a closed-cell structure, the rectangular block including a first face, a second face opposing the first face, and four side faces extending between the first face and the second face; supporting, by the rectangular block, a lower end of a sound projection device at an angle relative to a surface, the rectangular block providing acoustic isolation and/or vibration dampening between the sound projection device and the surface; and positioning the rectangular block between a proximal and a distal edge of the lower end of the sound projection device to selectively adjust the angle of the sound projection device relative to the surface, thereby modifying a direction of sound projection. At least two of the side faces have dimensions varying in height, width, or length.

[0016] In another aspect of the present disclosure, the method may further include supporting, by the rectangular block, a neck of a musical instrument above the surface during maintenance of the musical instrument.

[0017] In yet another aspect of the present disclosure, the method may further include securing, by the rectangular block, a position of a musical instrument with a container during storage and/or transportation of the musical instrument.

[0018] In a further aspect of the present disclosure, the method may further include providing, by the rectangular block, a space between a plurality of musical instruments stored on a stand and/or rack.

[0019] In yet a further aspect of the present disclosure, the method may further include fabricating the rectangular block from cross-linked polyethylene (XLPE), chemically cross-linked polyethylene (XPE), and/or ethylene vinyl acetate (EVA).

[0020] In another aspect of the present disclosure, the rectangular block may have a length ranging from approximately 2 inches to 6 inches, a width ranging from approximately 1.5 inches to 4 inches, and a height ranging from approximately 1 inch to 3 inches.

[0021] In yet another aspect of the present disclosure, the method may further include rotating the rectangular block about an axis relative to the surface to adjust a height of the rectangular block. The rectangular block may have a first orientation configured to tilt the sound projection device at a first angle relative to the surface, and a second orientation configured to tilt the sound projection device at a second angle relative to the surface.

[0022] In a further aspect of the present disclosure, the first angle may range from approximately 15 degrees to 31 degrees, and the second angle may range from approximately 25 degrees to 33 degrees.

[0023] In yet a further aspect of the present disclosure, the rectangular block may further include at

least one arcuate recess defined in a first side face thereof. The arcuate recess may be configured to accommodate a neck of a musical instrument.

[0024] In accordance with an aspect of the present disclosure, a multi-use apparatus includes an asymmetrical block fabricated from a foam having a closed-cell structure. The asymmetrical block includes a first face, a second face opposing the first face, and four side faces extending between the first face and the second face. At least two of the side faces have dimensions varying in height, width, or length. The asymmetrical block is configured to support a lower end of a sound projection device at an angle relative to a surface, the asymmetrical block providing acoustic isolation and/or vibration dampening between the sound projection device and the surface. Positioning the asymmetrical block between a proximal and a distal edge of the lower end of the sound projection device selectively adjusts the angle of the sound projection device relative to the surface, thereby modifying a direction of sound projection.

[0025] Further details and aspects of the present disclosure are described in more detail below with reference to the appended figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] A better understanding of the features and advantages of the present disclosure will be obtained by reference to the following detailed description that sets forth illustrative aspects, in which the principles of the present disclosure are utilized, and the accompanying figures of which:

[0027] FIG. 1 is a perspective view of a multi-use apparatus, in accordance with aspects of the present disclosure;

[0028] FIGS. 2A-2F are perspective views illustrating various configurations of a sound projection device and the multi-use apparatus of FIG. 1, in accordance with aspects of the present disclosure;

[0029] FIGS. 3A and 3B are perspective views illustrating configurations of a musical instrument and the multi-use apparatus of FIG. 1, in accordance with aspects of the present disclosure;

[0030] FIG. 4 is a perspective view illustrating a configuration of a stand, multiple musical instruments, and the multi-use apparatus of FIG. 1, in accordance with aspects of the present disclosure;

[0031] FIGS. 5A and 5B are perspective views illustrating configurations of a case, a musical instrument, and the multi-use apparatus of FIG. 1, in accordance with aspects of the present disclosure; and

[0032] FIG. 6 a perspective view of another multi-use apparatus, in accordance with aspects of the present disclosure; and

[0033] FIG. 7 is a block diagram of a method of using the multi-use apparatus of FIG. 1, in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

[0034] The present disclosure relates to a multi-use apparatus, and, more specifically, to a multi-use apparatus configured for use with audio equipment, which integrates angled support, vibration dampening, acoustic isolation, security, and/or storage functions into a single, versatile multi-use apparatus.

[0035] Aspects of the present disclosure are described in detail with reference to the figures wherein like reference numerals identify similar or identical elements.

[0036] Although the present disclosure will be described in terms of specific aspects and examples, it will be readily apparent to those skilled in this art that various modifications, rearrangements, and substitutions may be made without departing from the spirit of the present disclosure. The scope of the present disclosure is defined by the claims appended hereto.

[0037] For purposes of promoting an understanding of the principles of the present disclosure,

reference will now be made to exemplary aspects illustrated in the figures, and specific language will be used to describe the same. It will nevertheless be understood that no limitation of the scope of the present disclosure is thereby intended. Any alterations and further modifications of the novel features illustrated herein, and any additional applications of the principles of the present disclosure as illustrated herein, which would occur to one skilled in the relevant art and having possession of this disclosure, are to be considered within the scope of the present disclosure.

[0038] As used herein, “audio equipment” may include any equipment and/or device any equipment used in the music industry, including devices for producing, amplifying, projecting, and/or manipulating sound, such as speaker cabinets, audio amplifiers, PA systems, studio monitors, sound reinforcement systems, and/or other electronic or acoustic sound-emitting devices. In addition, audio equipment may include various speaker configurations, such as guitar amplifier cabinets, bass amplifier cabinets, combo amplifiers, 4×12 and 2×12 speaker cabinets, and/or other enclosures housing one or more speakers for sound projection. Further, audio equipment may extend to devices used in live performances, recording studios, rehearsal spaces, and/or general sound reinforcement applications, ensuring broad applicability across musical, commercial, and professional audio settings.

[0039] The multi-use apparatus described herein addresses a range of common challenges faced by musicians, audio engineers, sound technicians, and other users, enabling more efficient transport, setup, and handling of audio equipment. Using state-of-the-art compositions, the lightweight construction avoids unnecessary bulk while still maintaining sufficient durability and functionality for professional use. In addition, the dimensions allow for an optimized tilt angle for sound projection devices and more ergonomic and/or secure positing during maintenance. Unlike bulkier repair stands or dedicated neck cradles, the multi-use apparatus is compact and lightweight, allowing musicians and technicians to easily transport it in a gig bag, case, or tool kit. This portability makes the multi-use apparatus particularly useful for on-the-go maintenance, whether in a touring environment, recording studio, rehearsal space, or backstage setting. While this disclosure focuses on a multi-use apparatus for use with a sound projection device and musical instruments, the multi-use apparatus may be adapted for a broad range of additional applications and/or devices. In particular, the adaptable configuration and supportive properties of the multi-use apparatus enable application for any scenario requiring stable positioning, adjustable angling, acoustic isolation, vibration dampening, and/or secure storage.

[0040] Referring to FIG. 1, a multi-use apparatus **100** is shown according to aspects of the present disclosure. Multi-use apparatus **100** generally includes side faces **102**, **104**, **106**, **108**, first face **110**, and second face **112**, e.g., a six-sided block-like structure (e.g., a rectangular block). The first face **110** (e.g., an upper face) and the second face **112** (e.g., a lower face) are opposing faces that define the uppermost and lowermost surfaces, respectively, of the multi-use apparatus **100**. The side faces **102**, **104**, **106**, **108** extend between the first face **110** and the second face **112**, defining the lateral boundary of the multi-use apparatus **100**, e.g., forming the side surfaces of the multi-use apparatus **100** and/or enclosing the block-like structure. Opposing pairs of side faces may include side faces **102**, **106** and side faces **104**, **108**, which are positioned opposite each other.

[0041] Side faces **102**, **104**, **106**, **108** may extend along a vertical axis (e.g., axis H), while the first face **110** and the second face **112** may extend along a horizontal axis (e.g., axis W), which is perpendicular to the vertical axis (H). While generally described as a six-sided block, the multi-use apparatus **100** may be configured with any number of suitable shapes, faces, and/or dimensions to accommodate different applications of the multi-use apparatus **100** (e.g., sound projection device, spacer, and/or musical instrument support system). In aspects, the multi-use apparatus **100** may be fabricated as a block, a cube, a rectangle, or a combination thereof with adjustable configurations.

[0042] In aspects, at least one face of multi-use apparatus **100** may be inclined at a non-perpendicular angle, such that at least one side face **102**, **104**, **106**, **108** defines an acute and/or obtuse angle relative to an adjacent side face, creating a non-uniform profile of the multi-use

apparatus **100**. For example, at least one side face **102**, **104**, **106**, **108** may be inclined at a 45° angle relative to an axis (e.g., axis L or axis W) defined by the first face **110** and the second face **112**, forming an irregular and/or asymmetrical shape while maintaining the overall six-sided structure. This wedge-like shape of a six-sided structure may provide an optimized tilt angle for sound projection devices, (e.g., improving sound projection and acoustic isolation), enable more ergonomic and/or secure posing (e.g., when used as a support surface for a musical instrument), and/or enable additional height and tilt configurations for adaptability in various storage and/or spacing applications.

[0043] The multi-use apparatus **100** is generally configured as a block with dimensions of approximately 4 inches in length, 3 inches in width, and 2 inches in height (e.g., 4"L×3"W×2"D block). Generally, the dimensions of the multi-use apparatus **100** range between approximately 3-6 inches in length, 1.5-4 inches in width, and 1-3 inches in height. Such compact dimensions optimize the functionality of the multi-use apparatus **100** for various applications, thereby facilitating the ease of setup, transportation, and/or storage thereof. For example, the dimensions ensure compatibility with a wide range of sound projection devices (FIGS. 2A-2F), providing stable and/or adjustable tilt angles while maintaining effective acoustic isolation. Additionally, the dimensions allows the apparatus to function as a support for musical instruments and as a spacer within instrument cases or storage setups.

[0044] Generally, at least two of the side faces **102**, **104**, **106**, **108** vary in dimension, e.g. vary in at least one of length, width, or height. This difference in dimensions allows the multi-use apparatus **100** to be positioned in multiple orientations, enabling various functional advantages such as tilting variations (FIGS. 2A-2F), instrument support adaptability (FIGS. 3A, 3B), and/or storage or spacer functionality (FIGS. 4, 5A, 5B). For example, the first face **110** and the second face **112** of the multi-use apparatus **100** may be configured with the same dimensions. For example, the first face **110** and the second face **112** may each measure approximately 4 inches in length and 3 inches in width, corresponding to the overall length and width of the multi-use apparatus **100**. In aspects, at least two of the side faces **102**, **104**, **106**, **108** are configured to vary in dimension. For example, side faces **102**, **106** may each measure approximately 3 inches in width and 2 inches in height, corresponding to the overall height and width of the multi-use support apparatus. In another example, side faces **104**, **108** may each measure approximately 4 inches in length and 2 inches in height, corresponding to the overall height and length of the multi-use apparatus **100**.

[0045] The multi-use apparatus **100** is generally fabricated from cross-linked polyethylene (XLPE) foam, which provides an optimal balance of durability and/or noise reduction while maintaining a lightweight structure for ease of transport and repositioning. XLPE provides various advantages over traditional foam applications, particularly superior sound absorption and/or acoustic isolation properties. The dense, closed-cell structure of the XLPE foam is also configured to provide heightened shock absorption and enhance a resistance of the multi-use apparatus **100** to compression, moisture, chemicals, temperature, and/or fire, enabling ease of repeated use in supporting sound projection devices, musical instruments, and/or other equipment. In addition, XLPE is eco-friendly as it requires less energy to produce and is highly durable, thereby reducing waste over time.

[0046] In aspects, the multi-use apparatus **100** may incorporate materials such as chemically cross-linked polyethylene (XPE), ethylene vinyl acetate (EVA) foam, polyvinyl chloride foam (PVC) foam, expanded polyethylene (EPE) foam, ethylene propylene diene monomer (EPDM) foam, open cell, filter foam, natural rubber, polyisocyanurate foam, and/or rigid polymers, based on a desired level of firmness, flexibility, and/or impact resistance for a particular application. For example, EVA foam is another closed-cell structure, which has a generally lighter weight and provides advantages including low water absorption, ultraviolet radiation (UV) resistance, high durability, enhanced flexibility and cushioning, and/or stress-crack resistance. For example, a layer of EVA foam may be incorporated in the multi-use apparatus **100** to enhance a flexibility for ease of

molding into different shapes and sizes while maintaining structural integrity (e.g., to conform to a contour within an instrument case). EVA foam may also enhance a cushioning property of the multi-use apparatus **100** where impact absorption is critical (e.g., shock absorption within an instrument case). For example, a layer of rubber composites and/or other polymers may provide enhanced grip and/or impact resistance.

[0047] In aspects, a combination of foam and/or polymers may be incorporated into multi-use apparatus **100** to achieve specific acoustic damping properties, thereby ensuring effective minimization of vibrations and/or unwanted resonance when the multi-use apparatus **100** is used as a support or spacer. For example, a layered and/or composite configuration (e.g., a dual-layered structure) combining XLPE for structural integrity with EVA for additional cushioning may optimize acoustic isolation. In another example, an external PVC-coated layer and/or textured surface may be applied to improve durability and surface grip, reducing movement on a smooth or hard surface **210**. These combinations allow the multi-use apparatus **100** to effectively function across multiple applications, including supporting cabinets, stabilizing musical instruments during maintenance, and/or securing instruments within cases during transport.

[0048] Now referring to FIGS. 2A-2F various configurations of a sound projection device **200** and the multi-use apparatus **100** are shown, according to aspects of the present disclosure. In aspects, sound projection device **200** is a 1×12" speaker. As used herein, a sound projection device **200** may include a speaker amplifier cabinet, a guitar amplifier, a combination amplifier, a 4×12 speaker cabinet, a 2×12 speaker cabinet, a bass amplifier cabinet, a keyboard amplifier, and/or any other enclosure housing one or more speakers for sound amplification.

[0049] The multi-use apparatus **100** is configured to support a lower end **202** of the sound projection device **200**. In doing so, the multi-use apparatus **100** serves a dual function by enabling adjustable tilt positioning and acoustic isolation and/or vibration dampening for the sound projection device **200**. For example, the multi-use apparatus **100** is configured to enable adjustable tilt positioning of the sound projection device **200** at an angle relative to a surface **210** (e.g., a stage, platform, riser, and/or other surface) along an axis X. The angle may be adjusted in a desired direction to influence a sound projection **220**, e.g., a projection of sound waves toward a musician and/or an audience. In addition, the multi-use apparatus **100** is configured to enhance acoustic and/or harmonic isolation properties of the sound projection device **200** by reducing vibration transfer between the sound projection device **200** and the surface **210**. The multi-use apparatus **100** provides an advantage over traditional tilt and/or support devices by enabling a greater range of tilt and lesser degrees in such tilt, while preventing the sound projection device **200** from falling, e.g., preventing an amplifier cabinet from falling to a base.

[0050] The multi-use apparatus **100** is generally placed below a central point of the lower end **202** of the sound projection device **200** for optimal sound absorption and/or acoustic isolation.

However, the multi-use apparatus **100** may be placed in any number of depth configurations, such as anywhere along the lower end **202** of the sound projection device **200** between a proximal edge **202a** and a distal edge **202b**. In addition, the multi-use apparatus **100** can be configured in various orientations based on which face of the multi-use apparatus **100** is supporting the lower end **202** of the sound projection device **200**. By rotating the multi-use apparatus **100** about an axis (e.g., X) relative to the surface **210**, an alternative orientation of apparatus **100** may be achieved. The depth configuration and/or the orientation of multi-use apparatus **100** may determine the tilt positioning of the sound projection device **200**, defining an adjustable angle of the sound projection device **200** along the axis X relative to the surface **210**. For example, a configuration closer to proximal edge **202a** of sound projection device **200** and/or a shorter orientation of multi-use apparatus **100** may result in a lower angle, whereas a configuration closer to distal edge **202b** of sound projection device **200** and/or a taller orientation of multi-use apparatus **100** may result in a higher angle.

[0051] With further reference to FIGS. 2A-2C, configurations of the sound projection device **200** and the multi-use apparatus **100** in the first orientation O1 (e.g., a "standard" placement) are

shown. In the first orientation O1, the first face **110** of the multi-use apparatus **100** may be supporting the lower end **202** of the sound projection device **200**, defining a 4"×3"×2" rectangular block. As the depth configuration of the multi-use apparatus **100** moves closer to the distal edge **202b** of the sound projection device **200**, the angle of the sound projection device **200** may increase, ranging between approximately 15°-31° vertical tilt (e.g., with a multi-use apparatus **100** based on a 4"×3"×2" 4-pound XLPE foam block and a sound projection device **200** against a flat surface **210**). For example, FIG. 2A illustrates the multi-use apparatus **100** in a first configuration, e.g., extending approximately 1" outward and/or proximally from the proximal edge **202a** to define a first angle A (e.g., 15°). FIG. 2B illustrates the multi-use apparatus **100** in a second configuration, e.g., flush against proximal edge **202a**, to define a second angle A' (e.g., 20°). FIG. 2C illustrates the multi-use apparatus **100** in a third configuration, e.g., extending approximately 1" inward and/or distally from the inner edge **202a** to define a third angle A" (e.g., 31°).

[0052] With further reference to FIGS. 2D-2F, configurations of the sound projection device **200** and the multi-use apparatus **100** in a second orientation O2 (e.g., a "tall side orientation") are shown. In the second orientation O2, a side face **102** of the multi-use apparatus **100** may be supporting the lower end **202** of the sound projection device **200**, defining a 3"×2"4" rectangular block. As the depth configuration of the multi-use apparatus **100** moves closer to the distal edge **202b** of the sound projection device **200**, the angle of the sound projection device **200** may increase, ranging between approximately 25°-33° vertical tilt. For example, FIG. 2D illustrates the multi-use apparatus **100** in a fourth configuration, e.g., extending approximately 1" outward and/or proximally from the proximal edge **202a** to define a first angle a (e.g., 25°). FIG. 2E illustrates the multi-use apparatus **100** in a fifth configuration, e.g., flush against the proximal edge **202a**, to define a second angle A' (e.g., 27°). FIG. 2F illustrates the multi-use apparatus **100** in a sixth configuration, e.g., extending approximately 1" inward and/or distally from the proximal edge **202a** to define a third angle A" (e.g., 33°).

[0053] In aspects, in a third orientation O3, a side face **108** of the multi-use apparatus **100** may be supporting the lower end **202** of the sound projection device **200**, defining a 4"×2"3" rectangular block (e.g., a "third height"). This and other orientations and/or configurations are contemplated and understood by one of ordinary skill in the art. For example, various side-to-side configurations of the multi-use apparatus **100** while supporting the lower end **202** of the sound projection device **200** are contemplated.

[0054] Now referring to FIGS. 3A and 3B, configurations of a musical instrument **300** and the multi-use apparatus **100** are shown, according to aspects of the present disclosure. In aspects, the musical instrument **300** is a string instrument. As used herein, a musical instrument **300** may include any device and/or apparatus designed to produce sound for musical performance, recording, and/or amplification, including string instruments (e.g., guitars, bass guitars, violins, cellos), percussion instruments (e.g., drums, electronic drum pads), wind instruments (e.g., saxophones, trumpets, flutes), keyboard instruments (e.g., pianos, synthesizers, electric keyboards), and/or electronic or digital instruments (e.g., musical instrument digital interface (MIDI) controllers, samplers, and other electronically generated sound devices).

[0055] A composition of the multi-use apparatus **100** (e.g., XLPE and/or XPE) offers both structural support and compressibility to cushion the musical instrument **300**. The closed-cell structure of the multi-use apparatus **100** provides resilience and durability while offering sufficient support to prevent unwanted movement of a musical instrument **300**. This composition ensures that the multi-use apparatus **100** maintains a shape under load while also absorbing minor vibrations and/or impact forces that may transfer to the musical instrument **300**. For example, based on a 4"×3"×2" 4-pound XLPE foam block with an average compressive strength of 25 pounds per square inch (PSI), the multi-use apparatus **100** may support up to 300 pounds of weight. Therefore, the XLPE foam supports an enhanced amount of weight for a relatively small sized device, thereby enabling support for a variety of sizes and shapes of musical instruments **300**. In addition, the non-

slip properties of XLPE and/or XPE foam further enhance grip and adherence of the multi-use apparatus **100** to the surface **304**, thus preventing unintended movement when used on smooth surfaces **304** such as workbenches and/or stage floors.

[0056] With further reference to FIGS. **3A** and **3B**, multi-use apparatus **100** is configured to support a neck **302** of the musical instrument **300** by providing a sturdy yet flexible resting surface, which supports and/or elevates the neck **302** of the musical instrument **300** above a working surface **310**. For example, a user may rest the neck **302** of the musical instrument **300** on the first surface **110** of the multi-use apparatus **100** to enable ease of changing strings, adjusting hardware, tuning, cleaning, and/or other maintenance tasks related to the musical instrument **300**. Similar to FIGS. **2A-2F**, the multi-use apparatus **100** can be positioned in multiple orientations to accommodate musical instruments of diverse sizes. By adjusting the orientation of the multi-use apparatus **100**, a height at which the neck **302** of the musical instrument **300** is supported can be selectively adjusted, providing versatility for different maintenance needs and/or musical instruments **300** with varying neck profiles and thicknesses. In addition, a face (e.g., a flat second face **112**) of the multi-use apparatus **100** provides a stable foundation, minimizing tipping and/or instability when supporting the musical instrument **300** during maintenance tasks.

[0057] For example, FIG. **3A** illustrates the multi-use apparatus **100** in a first orientation **O1** (e.g., a “standard” placement). In the first orientation **O1**, the first face **110** of the multi-use apparatus **100** may be supporting the neck **302** of the musical instrument **300**, defining a 4”×3”×2” rectangular block. FIG. **3B** illustrates the multi-use apparatus **100** in a second orientation **O2** (e.g., a “tall side orientation”). In the second orientation **O2**, a side face **102** of the multi-use apparatus **100** may be supporting the neck **302** of the musical instrument **300**, defining a 3”×2”×4” rectangular block. In aspects, a third orientation may be configured. The adjustable orientation of multi-use apparatus **100** allows for varied configurations, making the multi-use apparatus **100** adaptable for use with various musical instrument **300s**.

[0058] Now referring to FIG. **4**, a configuration of a rack **400**, musical instruments **300**, and the multi-use apparatus **100** is shown, according to aspects of the present disclosure. The multi-use apparatus **100** is configured to act as a spacer, e.g., being positioned between multiple musical instruments **300** while placed on the rack **400** as to maintain separation and direction contact among the musical instruments **300** during storage. The rack **400** may include a stand, a storage rack, a display system, and/or any other structure configured to hold multiple musical instruments **300**.

[0059] For example, multiple multi-use apparatuses **100** may be placed firmly between the necks **302** of each musical instrument **300** on the rack **400**, thereby minimizing the risk of scratches, dents, or other damage that may occur from accidental impact. The closed-cell foam composition (e.g., XLPE, XPE, and/or EVA) provides a soft and durable cushion to absorb minor vibrations and/or maintaining structural integrity of each multi-use apparatus **100** under pressure. Additionally, the varying side configurations and/or orientations (e.g., **O1**, **O2**, **O3**) of the multi-use apparatus **100** enable adjustable spacing, accommodating varied sizes of musical instruments **300** and provide extra cushioning (e.g., **O2** may provide additional protection) to enable a flexible arrangement of musical instruments **300** on the rack **400**. This functionality provides an advantage of multi-use apparatus over traditional storage devices, particularly in studio, retail, and/or touring environments where musical instruments **300** are stored closely together and require organized, protective separation to maintain optimal condition.

[0060] Now referring to FIGS. **5A** and **5B**, configurations of a case **500**, a musical instrument **300**, and the multi-use apparatus **100** are shown. The multi-use apparatus **100** is configured to act as a spacer and/or guard, e.g., being positioned between a musical instruments **300** and a case wall **502** of a case **500** while stored therein to secure the musical instrument **300** in a position and maintain separation and direction contact with the case **500** during transportation and/or storage. The case **500** may include soft cases, fitted cases, and/or hard-shell cases, each of which presents unique

challenges in ensuring a proper fit and preventing instrument movement.

[0061] FIG. 5A illustrates a musical instrument **300** stored in a case **500** having a soft configuration. In the soft configuration of the case **500**, there may be no internal shape and/or specific containment mechanism within the case **500** to secure and/or stabilize the musical instrument **300** therein, thereby providing a loose fit. The multi-use apparatus **100** may be used as a spacer and/or filler to reduce excess space within case **500** and/or prevent unwanted shifting of the musical instrument **300** when inside the case **500**. For example, the multi-use apparatus **100** may be placed in the first orientation O1, where side face **102** is disposed on the musical instrument **300** and side face **106** is disposed on the case wall **502**.

[0062] FIG. 5B illustrates a musical instrument **300** stored in a case **500** having a fitted configuration. In the fitted configuration of the case **500**, there may be an internal shape and/or specific containment mechanism designed for a specific musical instrument **300**, thereby providing a tight fit. The multi-use apparatus **100** may be used as a spacer and/or filler to reduce excess space within case **500**, which may be significantly less than a case **500** having a soft configuration. For example, the multi-use apparatus **100** may be placed in the second orientation O2, where first face **110** is disposed on the musical instrument **300** and second face **112** is disposed on the case wall **502**.

[0063] In aspects, case **500** may be a hard-shell case, which typically has a generic shape not specific to any musical instrument **300**. Therefore, similar to a soft configuration and/or a fitted configuration of case **500**, the multi-use apparatus **100** may be used as a spacer and/or filler to reduce excess space within case **500**. Moreover, due to the lack of cushion in the hard-shell configuration of case **500**, the closed-cell foam composition (e.g., XLPE or EVA) provides additional impact resistance, enhancing protection of the musical instrument **300** from external forces during transportation. Any orientation (O1, O2, O3) of the multi-use apparatus **100** may be used for flexibility in strategic placement inside case **500**.

[0064] Now referring to FIG. 6, a multi-use apparatus **600** is shown. The multi-use apparatus **600** generally includes side faces **602**, **604**, **606**, **608**, a first face **610**, and a second face **612**, e.g., a six-sided block-like structure (e.g., a rectangular block). The first face **610** (e.g., an upper face) and the second face **612** (e.g., a lower face) are opposing faces that define the uppermost and lowermost surfaces, respectively, of the multi-use apparatus **600**. The side faces **602**, **604**, **606**, **608** extend between the first face **610** and the second face **612**, defining the lateral boundary of the multi-use apparatus **600**, e.g., forming the side surfaces of the multi-use apparatus **600** and/or enclosing the block-like structure. Opposing pairs of side faces may include side faces **602**, **606** and side faces **604**, **608**, which are positioned opposite each other. The multi-use apparatus **600** discloses features similar to the multi-use apparatus **100**, with the exception of the features discussed below.

[0065] An arcuate recess **614** may be defined in a face of multi-use apparatus **600**. The arcuate recess **614** is configured to accommodate the neck **302** of the musical instrument **300**, thereby enhancing stabilization and/or securement of the musical instrument **300** during maintenance to prevent unintended movement thereof. For example, the arcuate recess **614** may be defined in a side face **102**, **104**, **106**, **108**, thereby confirming to the shape of a neck **302** of a musical instrument. The arcuate recess **614** may be configured with various depths. For example, the arcuate recess **614** may have a shallow curve configured to broaden a contact surface, thereby distributing weight more evenly across the neck **301** and/or body of the musical instrument. In another example, the arcuate recess **614** may have a deeper recess configured to cradle the neck **302** of the musical instrument **300** more securely, minimizing lateral movement during maintenance thereof. In aspects, the arcuate recess **614** may be used as a storage for equipment, for example, for holding strings during maintenance of the musical instrument **300**.

[0066] While described as an arc-shaped recess, arcuate recess **614** may be in any shape, size, and configuration. For example, the arcuate recess **614** may be an angled and/or stepped recess configured to provide multiple resting points depending on the instrument's neck shape. In aspects,

the multi-use apparatus **600** may include multiple arcuate recesses **614** positioned on different faces, allowing for adjustable height and support configurations. In addition, a grip **616** may be disposed on a face of multi-use apparatus **600**. The grip **616** is configured to enhance stability, surface adherence, and/or vibration isolation of the multi-use apparatus **600** to prevent unintended movement during use. For example, the grip **616** may be a polymer-based structure affixed to a face of the multi-use apparatus **600**. The grip **616** may be glued, heat-bonded, and/or molded onto the apparatus to ensure a durable, long-lasting attachment that withstands repeated use. In another example, the grip **616** may be a textured surface defined in the multi-use apparatus **600** and/or applied as a separate layer to increase traction. The textured surface of the grip **616** may include raised patterns, grooves, ridges, and/or micro-perforations, designed to provide grip in different orientations of use. The placement of grip **616** may vary based on an intended use. For example, when used with sound projection device **200**, grip **616** may be placed on second face **212** to enhance friction with the surface **210**. In another example, when used as a support for musical instrument **300**, grip **616** may be included within the arcuate recess **614** to prevent the neck **302** from shifting therein. In another example, when used as a spacer in a case **500**, grip **616** may be included on side faces **602**, **606** to enhance friction with case walls **502**.

[0067] FIG. 7 shows a method **700** for an exemplary use of the multi-use apparatus **100**, according to aspects of the present disclosure. Although the steps of method **700** of FIG. 7 are shown in a particular order, the steps need not all be performed in the specified order, and certain steps can be performed in another order. In various aspects, the method **700** of FIG. 7 may be performed all or in part by components of the multi-use apparatus **100**. These and other variations are contemplated to be within the scope of the present disclosure.

[0068] In step **702**, a user provides the multi-use apparatus **100**, **600**. For example, the user may provide an XLPE rectangular block (4"×3"×2") fabricated from a foam having a closed-cell structure. The multi-use apparatus may include a first face **110**, **610**, a second face **112**, **612**, and side faces **102**, **104**, **106**, **108**, **602**, **604**, **606**, **608**. The multi-use apparatus **100**, **600** may be placed on a surface **210**, **310**, between necks **302** of musical instruments **300**, and/or between a musical instrument **300** and a case **500**.

[0069] Next, at step **704**, the multi-use apparatus **100**, **600** supports a lower end **202** of a sound projection device **200** at an angle relative to a surface **210**. The multi-use apparatus **100**, **600** may provide at least one of acoustic isolation or vibration dampening between the sound projection device **200** and the surface **210**. For example, the multi-use apparatus **100**, **600** may be placed in a first orientation **O1** configured to tilt the sound projection device **200** at a first angle A, e.g., 15° relative to the surface **210**.

[0070] Next, at step **706**, the user may position the multi-use apparatus **100**, **600** between a proximal edge **202a** and a distal edge **202b** of the lower end **202** of the sound projection device **200**. For example, the multi-use apparatus **100**, **600** may be flush with a proximal edge **202a** of the lower end **202** of the sound projection device **200**, then pushed one inch distally towards the distal edge **202b** of the lower end **202**, thereby increasing the angle to a second angle A', e.g., 20° relative to the surface **210**.

[0071] The below is data from practical field testing, which provides a comparison of foam blocks having various compositions for acoustic isolation and/or vibration dampening. All subjects were tested at 90 decibels sound performance level (SPL) at 3 feet above a surface with a 4"×3"×2" block. Low and subharmonic frequencies are generally below 250 Hertz and more noticeably below 150 Hertz. The measurement and comparisons started at 220 Hertz, and swept downward to demonstrate the elimination of the most noticeable low frequencies.

XLPE Foam Block:

TABLE-US-00001 Feet Decibels Frequencies 220 Hertz and Below (to 20 Hertz) 1 3 3 4 6 6
Frequencies 110 Hertz and Below (to 20 Hertz) 1 4 3 5 6 7 Frequencies 75 Hertz and Below (to 20 Hertz) 1 5 3 6 6 9

Wooden Block:

TABLE-US-00002 Feet Decibels Frequencies 220 Hertz and Below (to 20 Hertz) 1 0.5 3 1 6 1
Frequencies 110 Hertz and Below (to 20 Hertz) 1 1 3 1.35 6 1.5 Frequencies 75 Hertz and Below
(to 20 Hertz) 1 1 3 1.75 6 2

Concrete Brick:

TABLE-US-00003 Feet Decibels Frequencies 220 Hertz and Below (to 20 Hertz) 1 0.75 3 1.25 6
1.5 Frequencies 110 Hertz and Below (to 20 Hertz) 1 0.9 3 1.15 6 1.35 Frequencies 75 Hertz and
Below (to 20 Hertz) 1 1 3 1.15 6 1.3

[0072] Conclusion: the XPE foam block provides substantially more acoustic isolation and reduces transference of sub-low harmonic frequencies between an amplifier cabinet and surface area (i.e., stage and/or platform) than traditional and/or generic blocks.

[0073] Certain aspects of the present disclosure may include some, all, or none of the above advantages and/or one or more other advantages readily apparent to those skilled in the art from the figures, descriptions, and claims included herein. Moreover, while specific advantages have been enumerated above, the various aspects of the present disclosure may include all, some, or none of the enumerated advantages and/or other advantages not specifically enumerated above.

[0074] The aspects disclosed herein are examples of the disclosure and may be embodied in various forms. For instance, although certain aspects herein are described as separate aspects, each of the aspects herein may be combined with one or more of the other aspects herein. Specific structural and functional details disclosed herein are not to be interpreted as limiting, but as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the present disclosure in virtually any appropriately detailed structure. Like reference numerals may refer to similar or identical elements throughout the description of the figures.

[0075] The phrases “in an embodiment,” “in aspects,” “in various aspects,” “in some aspects,” or “in other aspects” may each refer to one or more of the same or different example Aspects provided in the present disclosure. A phrase in the form “A or B” means “(A), (B), or (A and B).” A phrase in the form “at least one of A, B, or C” means “(A); (B); (C); (A and B); (A and C); (B and C); or (A, B, and C).”

[0076] It should be understood that the foregoing description is only illustrative of the present disclosure. Various alternatives and modifications can be devised by those skilled in the art without departing from the disclosure. Accordingly, the present disclosure is intended to embrace all such alternatives, modifications, and variances. The aspects described with reference to the attached figures are presented only to demonstrate certain examples of the disclosure. Other elements, steps, methods, and techniques that are insubstantially different from those described above and/or in the appended claims are also intended to be within the scope of the disclosure.

Claims

1. A multi-use apparatus, comprising: a rectangular block fabricated from a foam having a closed-cell structure, the rectangular block including: a first face; a second face opposing the first face; and four side faces extending between the first face and the second face, wherein at least two of the side faces have dimensions varying in height, width, or length; wherein the rectangular block is configured to support a lower end of a sound projection device at an angle relative to a surface, the rectangular block providing at least one of acoustic isolation or vibration dampening between the sound projection device and the surface, and wherein positioning the rectangular block between a proximal and a distal edge of the lower end of the sound projection device selectively adjusts the angle of the sound projection device relative to the surface, thereby modifying a direction of sound projection.

2. The multi-use apparatus of claim 1, wherein the rectangular block is further configured to support a neck of a musical instrument above the surface during maintenance of the musical

instrument.

3. The multi-use apparatus of claim 1, wherein the rectangular block is further configured to secure a position of a musical instrument within a case during at least one of storage or transportation of the musical instrument.

4. The multi-use apparatus of claim 1, wherein the rectangular block is further configured to provide a space between a plurality of musical instruments stored on at least one of a stand or a rack.

5. The multi-use apparatus of claim 1, wherein the rectangular block is fabricated from at least one of cross-linked polyethylene (XLPE), chemically cross-linked polyethylene (XPE), or ethylene vinyl acetate (EPA).

6. The multi-use apparatus of claim 1, wherein the rectangular block has a length ranging from approximately 2 inches to 6 inches, a width ranging from approximately 1.5 inches to 4 inches, and a height ranging from approximately 1 inch to 3 inches.

7. The multi-use apparatus of claim 1, wherein the rectangular block has a first orientation configured to tilt the sound projection device at a first angle relative to the surface, and a second orientation configured to tilt the sound projection device at a second angle relative to the surface, the second orientation obtained by rotating the rectangular block about an axis relative to the surface.

8. The multi-use apparatus of claim 7, wherein the first angle ranges from approximately 15 degrees to 31 degrees, and the second angle ranges from approximately 25 degrees to 33 degrees.

9. The multi-use apparatus of claim 1, wherein the rectangular block further includes at least one arcuate recess defined in a first side face thereof, the arcuate recess configured to accommodate a neck of a musical instrument.

10. The multi-use apparatus of claim 1, wherein the rectangular block further includes an external layer configured to enhance a grip between the rectangular block and the surface.

11. A method for using a multi-use apparatus, comprising: providing a rectangular block fabricated from a foam having a closed-cell structure, the rectangular block including: a first face; a second face opposing the first face; and four side faces extending between the first face and the second face, wherein at least two of the side faces have dimensions varying in height, width, or length; supporting, by the rectangular block, a lower end of a sound projection device at an angle relative to a surface, the rectangular block providing at least one of acoustic isolation or vibration dampening between the sound projection device and the surface; and positioning the rectangular block between a proximal and a distal edge of the lower end of the sound projection device to selectively adjust the angle of the sound projection device relative to the surface, thereby modifying a direction of sound projection.

12. The method of claim 11, further comprising: supporting, by the rectangular block, a neck of a musical instrument above the surface during maintenance of the musical instrument.

13. The method of claim 11, further comprising: securing, by the rectangular block, a position of a musical instrument with a container during at least one of storage or transportation of the musical instrument.

14. The method of claim 11, further comprising: providing, by the rectangular block, a space between a plurality of musical instruments stored on at least one of a stand or rack.

15. The method of claim 11, further comprising: fabricating the rectangular block from at least one of cross-linked polyethylene (XLPE), chemically cross-linked polyethylene (XPE), or ethylene vinyl acetate (EPA).

16. The method of claim 11, wherein the rectangular block has a length ranging from approximately 2 inches to 6 inches, a width ranging from approximately 1.5 inches to 4 inches, and a height ranging from approximately 1 inch to 3 inches.

17. The method of claim 11, further comprising: rotating the rectangular block about an axis relative to the surface to adjust a height of the rectangular block, wherein the rectangular block has

a first orientation configured to tilt the sound projection device at a first angle relative to the surface, and a second orientation configured to tilt the sound projection device at a second angle relative to the surface.

18. The method of claim 17, wherein the first angle ranges from approximately 15 degrees to 31 degrees, and the second angle ranges from approximately 25 degrees to 33 degrees.

19. The method of claim 11, wherein the rectangular block further includes at least one arcuate recess defined in a first side face thereof, the arcuate recess configured to accommodate a neck of a musical instrument.

20. A multi-use apparatus, comprising: an asymmetrical block fabricated from a foam having a closed-cell structure, the asymmetrical block including: a first face; a second face opposing the first face; and four side faces extending between the first face and the second face, wherein at least one of the side faces defines an acute angle relative to an adjacent side face such that the asymmetrical block has a non-uniform profile; wherein the asymmetrical block is configured to support a lower end of a sound projection device at an angle relative to a surface, the asymmetrical block providing at least one of acoustic isolation or vibration dampening between the sound projection device and the surface, and wherein positioning the asymmetrical block between a proximal and a distal edge of the lower end of the sound projection device selectively adjusts the angle of the sound projection device relative to the surface, thereby modifying a direction of sound projection.
